ECMWF
EUROPEAN CENTRE FOR MEDIUM-RANGE WEATHER FORECASTS

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MOOC
Machine Learning in Weather & Climate

TRANSCRIPT

Forecasting with ML

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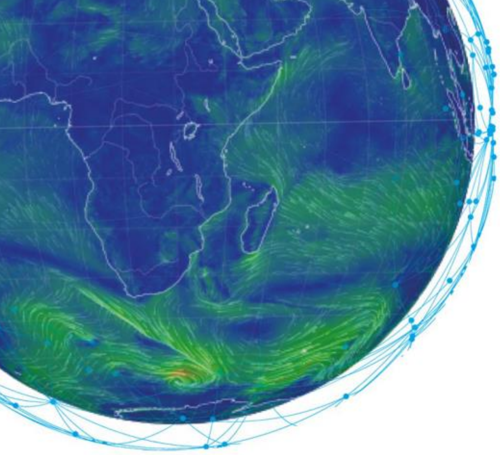

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Introduction

Hello and welcome to this introduction video to “Forecasting with Machine Learning”. My name is Lisa Burke and today we’ll meet Matthew Chantry, a machine learning scientist at ECMWF. With him, we’ll be exploring the many and varied ways machine learning can help in the forecast model.

So, how can machine learning help? Let’s go to ECMWF headquarters in Reading in the UK to discover it.

Script

Hello everyone. As we heard in previous sections, the forecast model is the heart of the prediction system, where an initial state of the atmosphere is evolved forward in time to give predictions for the earth system in the hours, weeks and even decades to come. Running this model is expensive, taking hours on some of the biggest super computers in the world to deliver a 10-day weather forecast.

Broadly we can divide machine learning approaches in the forecast model into two types, acceleration & model improvement.

In our first type we have efforts to accelerate components of the earth system model.

In a process often referred to as emulation, the aim is to create a machine learning emulator of an expensive component of an earth system model.

The most active area of research for this approach is in the parametrisation of physical processes that occur on smaller scales than the model grid.

This is motivated by the inherent uncertainty in these components and the fact that each vertical column of atmosphere is solved separately, creating a very GPU-friendly problem.

Why care about acceleration?

To reduce time to solution, meaning that a forecast could be delivered more quickly.

Alternately, this computational saving could be reinvested in improving some other aspect of the earth system model, for example, running a larger number of ensemble forecasts, a higher spatial resolution or increase model complexity. Any of these would improve the quality of the overall forecast.

The second approach to using machine learning in the forecasting model is using machine learning to directly improve the accuracy of the forecast model. This approach can leverage high resolution simulations or real-world observations to build new elements of the forecasting model.

Again, the physical parametrisation schemes are a natural target of this approach, because of their approximate nature.

Alternately, one could use machine learning to develop an online bias correction scheme. This scheme would aim to nudge the forecast at each timestep by compensation for predictable errors within the forecast model. This is very similar to an application of machine learning in the data assimilation task.

The most extreme example of machine learning in either of these contexts is in replacing the entire weather forecasting model.

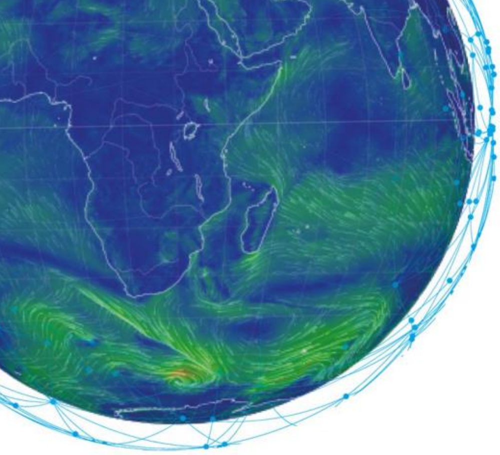
Several benchmark datasets for those interested in attempting this bold approach have been created, for example WeatherBench for medium-range forecasts. In a later module we'll take a deeper look at the opportunities and challenges when attempting such a problem.

In theory, given the right training data, a machine learning model of the entire weather forecasting system could be faster and more accurate.

However, currently machine learning models have yet to achieve this very challenging task.

In summary, we have seen that Machine Learning can help through a combination of acceleration and model improvement.

Despite being a very active area of research, as of recording, there have not yet been any deployed machine learning components within an operational forecasting model. This is partly due to the extremely high standards required to bring new elements into an operational forecasting model. Many simulations are required, ensuring that the stability and accuracy of the model are never compromised. However, it is likely in the years to come that this will change and we will see machine learning being deployed within the weather forecasting model. Whether we see machine learning replace the existing forecasting model will be left to a future discussion.



Summary

We’ve heard about two reasons to use machine learning in the context of the weather forecasting model. Both can lead to improvements in forecast quality.

Thank you, Matthew, for this great introduction on forecasting with Machine Learning. If you have any question or would like to discuss this subject in further detail, please use our forum on Moodle. It will be a pleasure to hear you!